

培道中學校

姓名：

(號)

成績

年級：

科目：

練習題 14.1b

1. (1) 記 $f(x, y) = 2x^2 - xy - y^2 - 6x - 3y + 5$

$$\frac{\partial f}{\partial x}(1, 2) = 4x - y - 6 \Big|_{(x, y) = (1, 2)} = 0 \quad \frac{\partial^2 f}{\partial x^2}(1, 2) = 4$$

$$\frac{\partial f}{\partial y}(1, 2) = -x - 2y - 3 \Big|_{(x, y) = (1, 2)} = 0 \quad \frac{\partial^2 f}{\partial y^2}(1, 2) = -2$$

$$\frac{\partial^2 f}{\partial x \partial y}(1, 2) = \frac{\partial^2 f}{\partial y \partial x}(1, 2) = -1 \quad \frac{\partial^n f}{\partial x \cdot \partial x \cdot \partial y} = 0 \quad (n \geq 3)$$

因此 $f(x, y) = f(1, 2) + \frac{1}{2}(x-1, y+2) \begin{pmatrix} 4 & -1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} x-1 \\ y+2 \end{pmatrix} + \dots$

$$= 5 + 2(x-1)^2 - (x-1)(y+2) - (y+2)^2$$

(2) 記 $f(x, y, z) = x^3 + y^3 + z^3 - 3xyz$

$$\frac{\partial f}{\partial x}(1, 1, 1) = 3x^2 - 3yz = 3(x^2 - yz) \Big|_{(x, y, z) = (1, 1, 1)} = 0 \quad \frac{\partial^2 f}{\partial x^2}(1, 1, 1) = 6x \Big|_{(x, y, z) = (1, 1, 1)} = 6$$

$$\frac{\partial^2 f}{\partial y \partial x}(1, 1, 1) = -3z \Big|_{(x, y, z) = (1, 1, 1)} = -3 \quad \frac{\partial^2 f}{\partial x^3} = 6$$

$$\frac{\partial^3 f}{\partial z \partial y \partial x} = -3 \quad \frac{\partial^3 f}{\partial y \partial x^2} = 0$$

因此 $f(x, y, z) = f(1, 1, 1) + \frac{1}{2}(x-1, y-1, z-1) \begin{pmatrix} 6 & -3 & -3 \\ -3 & 6 & -3 \\ -3 & -3 & 6 \end{pmatrix} \begin{pmatrix} x-1 \\ y-1 \\ z-1 \end{pmatrix} + \frac{1}{6} \times (6(x-1)^3 + 6(y-1)^3 + 6(z-1)^3 - 6(x-1)(y-1)(z-1))$

$$= 3[(x-1)^2 + (y-1)^2 + (z-1)^2] - (x-1)(y-1) - (y-1)(z-1) - (z-1)(x-1)$$

$$+ [(x-1)^3 + (y-1)^3 + (z-1)^3 - 3(x-1)(y-1)(z-1)]$$

$$= 3(x-1)^2 + 3(y-1)^2 + 3(z-1)^2 - 3(x-1)(y-1) - 3(y-1)(z-1) - 3(z-1)(x-1) +$$

$$(x-1)^3 + (y-1)^3 + (z-1)^3 - 3(x-1)(y-1)(z-1)$$

2.

$$f(x, y, z) = Ax^2 + By^2 + Cz^2 + 2Dxy + 2Eyz + 2Fzx$$

$$\begin{aligned} f(x+\Delta x, y+\Delta y, z+\Delta z) &= f(x, y, z) + (2Ax+2Dy+2Fz)\Delta x + (2By+2Ez+2Dx)\Delta y + (2Cz+2Fx+2Ey)\Delta z \\ &\quad + \frac{1}{2}(\Delta x \ \Delta y \ \Delta z) \begin{pmatrix} 2A & 2D & 2F \\ 2D & 2B & 2E \\ 2F & 2E & 2C \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \end{pmatrix} \\ &= f(x, y, z) + (2Ax+2Dy+2Fz)\Delta x + (2By+2Ez+2Dx)\Delta y + (2Cz+2Fx+2Ey)\Delta z \\ &\quad + A\Delta x^2 + B\Delta y^2 + C\Delta z^2 + 2D\Delta x\Delta y + 2E\Delta y\Delta z + 2F\Delta z\Delta x \end{aligned}$$

3. 記 $f(x, y) = x^y$

$$\frac{\partial f}{\partial x}(1, 1) = yx^{y-1} \Big|_{(x,y)=(1,1)} = 1, \quad \frac{\partial f}{\partial y}(1, 1) = x^y \ln x \Big|_{(x,y)=(1,1)} = 0$$

$$\frac{\partial^2 f}{\partial x^2}(1, 1) = y(y-1)x^{y-2} \Big|_{(x,y)=(1,1)} = 0, \quad \frac{\partial^2 f}{\partial y^2}(1, 1) = x^y (\ln x)^2 \Big|_{(x,y)=(1,1)} = 0$$

$$\frac{\partial^2 f}{\partial x \partial y}(1, 1) = x^y (\ln x)^2 + x^{y-1} \Big|_{(x,y)=(1,1)} = 1$$

$$\frac{\partial^2 f}{\partial x \partial y}(1, 1) \stackrel{!}{=} \frac{\partial^2 f}{\partial y \partial x}(1, 1) = 1$$

$$\begin{aligned} \Rightarrow x^y &= 1 + 1 \cdot (x-1) + \frac{1}{2}(x-1)(y-1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x-1 \\ y-1 \end{pmatrix} + o((x-1)^2 + (y-1)^2) \\ &= 1 + (x-1) + (x-1)(y-1) + o((x-1)^2 + (y-1)^2) \end{aligned}$$

4.

$$\frac{\partial}{\partial x} f(x, y) = \frac{\cos x}{\cos y}, \quad f(0, 0) = 1$$

$$\frac{\partial f}{\partial x}(0, 0) = -\frac{\sin x}{\cos y} \Big|_{(x,y)=(0,0)} = 0, \quad \frac{\partial f}{\partial y}(0, 0) = -\frac{\cos x}{\cos^2 y} \sin y \Big|_{(x,y)=(0,0)} = 0$$

$$\frac{\partial^2 f}{\partial y \partial x}(0, 0) = \frac{\partial^2 f}{\partial x \partial y}(0, 0) = -\frac{\sin x}{\cos^2 y} \sin y \Big|_{(0,0)} = 0$$

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2}(0, 0) &= -\frac{\cos x}{\cos y} \Big|_{(x,y)=(0,0)} = -1, \quad \frac{\partial^2 f}{\partial y^2}(0, 0) = -\cos x \cdot \frac{\cos^3 y - \sin y \cdot (2 \cos y \cdot (-\sin y))}{\cos^4 y} \Big|_{(x,y)=(0,0)} \\ &= -1 \end{aligned}$$

$$\Rightarrow \frac{\cos x}{\cos y} = 1 - \frac{1}{2}x^2 - \frac{1}{2}y^2 + o(x^2 + y^2) \approx 1 - \frac{1}{2}(x^2 + y^2)$$

姓名：

(號)

成績

年級：

科目：

練習題 14.11

$$1. (2) \quad Jf(x, y) = (2(1-3y)x \quad 3y^2-3x^2) = (0 \ 0)$$

$$\Rightarrow (x, y) = (0, 0) \text{ 或 } (\pm \frac{1}{3}, \frac{1}{3})$$

$$Hf(x, y) = \begin{pmatrix} 2(1-3y) & -6x \\ -6x & 6y \end{pmatrix} \Rightarrow Hf(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \text{ 不是正定矩陣, 也不是負定矩陣}$$

$$\forall \varepsilon > 0, \quad f(0, \varepsilon) = \varepsilon^3 > 0 = f(0, 0) > -\varepsilon^3 = f(0, -\varepsilon)$$

$\Rightarrow (0, 0)$ 不是 f 的極值點

$$Hf(\frac{1}{3}, \frac{1}{3}) = \begin{pmatrix} 0 & -2 \\ -2 & 2 \end{pmatrix}, \text{ 不是正定矩陣, 也不是負定矩陣}$$

$\Rightarrow (\frac{1}{3}, \frac{1}{3})$ 不是 f 的極值點

$$Hf(-\frac{1}{3}, \frac{1}{3}) = \begin{pmatrix} \frac{8}{3} & -2 \\ -2 & 2 \end{pmatrix} \Rightarrow \text{是正定矩陣} \Rightarrow (-\frac{1}{3}, \frac{1}{3}) \text{ 是 } f \text{ 的極小點, 極值是 } \frac{1}{27}$$

$$(4) \quad Jf(x, y) = (3x^2-3y \quad 3y^2-3x) = (0, 0) \Rightarrow \begin{cases} x^2=y \\ y^2=x \end{cases} \Rightarrow x^4=x \Rightarrow x=0 \text{ 或 } x=1$$

$$\Rightarrow (x, y) = (0, 0) \text{ 或 } (x, y) = (1, 1)$$

$$Hf(x, y) = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix} \Rightarrow Hf(0, 0) = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix} \text{ 是不定矩陣} \Rightarrow (0, 0) \text{ 不是 } f \text{ 的極值點}$$

$$Hf(1, 1) = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix} \text{ 是正定矩陣} \Rightarrow (1, 1) \text{ 是 } f \text{ 的極小點, 極值是 } -1$$

$$2. \quad |f(x, y)| = |x||y| \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \leq ab \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \left(\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \right)^2 \right)^{\frac{3}{2}} \cdot \frac{1}{3\sqrt{3}} = \frac{ab}{3\sqrt{3}}$$

(當且僅當 $\frac{x^2}{a^2} = \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}$ 時取等號)

$$\text{此時 } (x, y) = (\pm \frac{a}{\sqrt{3}}, \pm \frac{b}{\sqrt{3}}) \quad (x, y) = (\pm \frac{a}{\sqrt{3}}, \pm \frac{b}{\sqrt{3}}) \text{ 或 } (\mp \frac{a}{\sqrt{3}}, \pm \frac{b}{\sqrt{3}})$$

顯然當 $(x, y) = (\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$ 或 $(x, y) = (-\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 時, f 取最小值 $-\frac{ab}{3\sqrt{3}}$

當 $(x, y) = (-\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$ 或 $(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 時, f 取最大值 $\frac{ab}{3\sqrt{3}}$

因此 $(\mp \frac{a}{\sqrt{3}}, \pm \frac{b}{\sqrt{3}})$ 是極小點, $(\pm \frac{a}{\sqrt{3}}, \pm \frac{b}{\sqrt{3}})$ 是極大點

極小值是 $-\frac{ab}{3\sqrt{3}}$, 極大值是 $\frac{ab}{3\sqrt{3}}$

2. 記 $\frac{z^2}{c^2} = 1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}$ 且 $c > 0, z \geq 0$

$$f(x, y) = \frac{1}{c} xyz$$

$$Jf(x, y) = \frac{1}{c} \left(y(z + x \frac{\partial z}{\partial x}) - x(z + y \frac{\partial z}{\partial y}) \right) = (0, 0)$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \Rightarrow \begin{cases} \frac{2x}{a^2} + \frac{2z}{c^2} \cdot \frac{\partial z}{\partial x} = 0 \\ \frac{2y}{b^2} + \frac{2z}{c^2} \cdot \frac{\partial z}{\partial y} = 0 \end{cases} \Rightarrow Jf(x, y) = \frac{1}{c} \left(y(z - x \cdot \frac{x}{z} \cdot \frac{c^2}{a^2}) - x(z - y \cdot \frac{y}{z} \cdot \frac{c^2}{b^2}) \right) = (0, 0)$$

(i) 當 $z \neq 0$ 時, 只有 $(0, 0)$

3. $Jf(x, y) = (\cos x - \sin(x-y) - \sin y + \sin(x-y)) = (0, 0)$

$$\begin{cases} \cos x = \sin(x-y) \\ \sin y = \sin(x-y) \end{cases} \Leftrightarrow \begin{cases} \cos x = \sin x \cos y - \cos x \sin y \\ \cos x = \sin y \end{cases}$$

$$\Leftrightarrow \begin{cases} \cos^2 x + \cos x = \sin x \cos y \\ x + y = \frac{\pi}{2} \end{cases} \Rightarrow \cos^2 x + \cos x = \sin^2 x$$

$$\Leftrightarrow 2\cos^2 x + \cos x - 1 = 0 \Leftrightarrow \cos x = \frac{1}{2} \Leftrightarrow x = \frac{\pi}{3} \Leftrightarrow y = \frac{\pi}{6}$$

$$Hf(x, y) = \begin{pmatrix} -\sin x - \cos(x-y) & \cos(x-y) \\ \cos(x-y) & -\cos y - \cos(x-y) \end{pmatrix}$$

$$Hf\left(\frac{\pi}{3}, \frac{\pi}{6}\right) = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -(\frac{1+\sqrt{3}}{2}) \end{pmatrix} \text{ 是負定矩陣 } \Rightarrow \left(\frac{\pi}{3}, \frac{\pi}{6}\right) \text{ 是 } f \text{ 的極大點, 極大值是 } \frac{\sqrt{3}}{2}$$

4. 證: $\forall k \in \mathbb{R}$, 設 $y = kx$

$$f(x, kx) = x^2(3x^2 - 4kx + k^2) = 3x^4 - 4kx^3 + k^2x^2 = g(x)$$

$$\frac{dg}{dx} = 12x^3 - 12kx^2 - 2k^2x \Rightarrow \frac{dg(0)}{dx} = 0$$

$$\frac{d^2g}{dx^2} = 36x^2 - 24kx - 2k^2 \Rightarrow \frac{d^2g(0)}{dx^2} = -2k^2 \leq 0$$

當 $k \neq 0$ 時, $\frac{d^2g(0)}{dx^2} < 0 \Rightarrow$ 原點是極小點

當 $k = 0$ 時, $g(x) = 3x^4 \geq 0$ (當且僅當 $x = 0$ 時取等號) $\Rightarrow$ 原點是極小點

姓名：

(號)

成績

年級：

科目：

4 考慮直線 $x=0$.

$f(0, y) = y^2 \geq 0$ (當且僅當 $y=0$ 時取等號) $\Rightarrow$ 原點是極小點

$$f(x, y) = 3x^4 - 4x^2y + y^2 = (3x^2 - y)(x^2 - y)$$

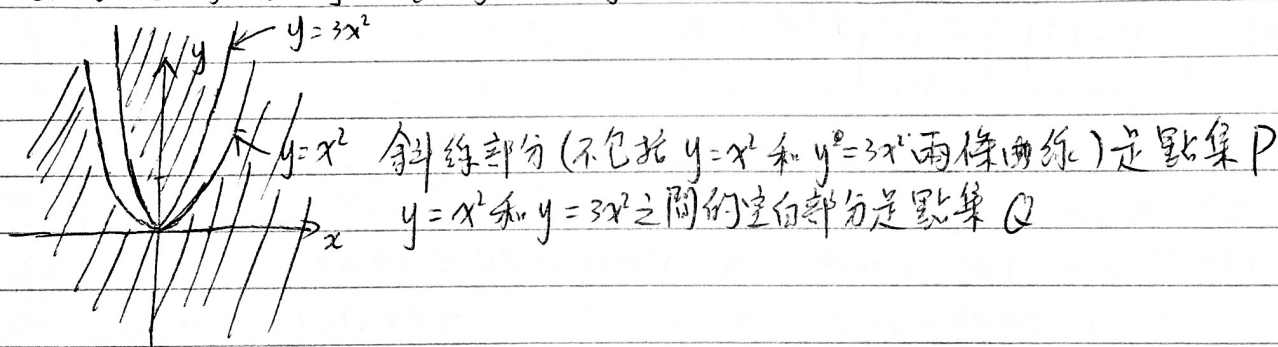
注意到, 當 $y = 2x^2$ 時, $f(x, y) \leq 0$.

因此, $\forall \varepsilon > 0$, $f(0, \varepsilon) = \varepsilon^2 > 0 = f(0, 0) > f(\varepsilon, 2\varepsilon^2)$

$\Rightarrow$ 原點不是極小點

$$P = \{(x, y) \mid f(x, y) > 0\} = \{(x, y) \mid y < x^2 \text{ 或 } y > 3x^2\}$$

$$Q = \{(x, y) \mid f(x, y) < 0\} = \{(x, y) \mid x^2 < y < 3x^2\}$$



練習題 14.12

1. (3) 記 $L(x, y, z, \lambda) = x - 2y + 2z + \lambda(x^2 + y^2 + z^2 - 1)$

$$(1 + 2\lambda x, -2 + 2\lambda y, 2 + 2\lambda z) = (0, 0, 0)$$

$$\Rightarrow x = -\frac{1}{2\lambda}, y = \frac{1}{\lambda}, z = -\frac{1}{\lambda}$$

$$\Rightarrow \left(-\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{\lambda}\right)^2 + \left(-\frac{1}{\lambda}\right)^2 = 1 \Leftrightarrow \lambda = \pm \frac{3}{2}$$

$$H_{L\lambda}(x, y, z, \lambda) = \begin{pmatrix} 2\lambda & 0 & 0 \\ 0 & 2\lambda & 0 \\ 0 & 0 & 2\lambda \end{pmatrix} \Rightarrow \begin{cases} \text{當 } \lambda = -\frac{3}{2} \text{ 時, } H_{L\lambda} \text{ 是負定矩陣, } \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right) \text{ 是極大點} \\ \text{極大值是 } 3 \\ \text{當 } \lambda = \frac{3}{2} \text{ 時, } H_{L\lambda} \text{ 是正定矩陣, } \left(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) \text{ 是極小點} \\ \text{極小值是 } -3 \end{cases}$$

(4) 記 $L(x, y, z, \lambda) = 3x^2 + 3y^2 + z^2 + \lambda(x + y + z - 1)$

$$\Rightarrow (6x + \lambda \quad 6y + \lambda \quad 2z + \lambda) = (0 \ 0 \ 0)$$

$$\Rightarrow x = -\frac{\lambda}{6}, y = -\frac{\lambda}{6}, z = -\frac{\lambda}{2} \Rightarrow -\frac{\lambda}{6} - \frac{\lambda}{6} - \frac{\lambda}{2} = 1 \Leftrightarrow \lambda = -\frac{6}{5}$$

$$\Leftrightarrow x = y = \frac{1}{5}, z = \frac{3}{5}$$

H.L. $\bar{x} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ 是正定矩陣, 因此 $(\frac{1}{5}, \frac{1}{5}, \frac{3}{5})$ 是極小點, 極小值是 $\frac{3}{5}$

2. 記 $u(x, y, z) = z^2$, $L(x, y, z, \lambda, \mu) = z^2 + \lambda(x^2 + y^2 - 2az) + \mu(x^2 + y^2 + xy - a^2)$

$$\Rightarrow (2(\lambda + \mu)x + \mu y \quad 2(\lambda + \mu)y + \mu x \quad 2z - 2a\lambda) = (0 \ 0 \ 0)$$

由 $x^2 + y^2 + xy = a^2 > 0$ 知 $\begin{pmatrix} x \\ y \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow z \neq 0 \Rightarrow \lambda = \frac{z}{a} \neq 0$

因此 $\begin{pmatrix} 2(\lambda + \mu) \\ \mu \end{pmatrix}, \begin{pmatrix} -\mu \\ 2(\lambda + \mu) \end{pmatrix}$ 线性相關

$$\Leftrightarrow [2(\lambda + \mu) - \mu][2(\lambda + \mu) + \mu] = 0$$

$$\Leftrightarrow \mu = -2\lambda \text{ 或 } \mu = -\frac{2}{3}\lambda$$

$$\Rightarrow z = a \Rightarrow (\mu, \lambda) = (-2, 1)$$

因此當 $\mu = -2\lambda$ 時, $x = -y \Rightarrow x^2 = y^2 = a^2 \Rightarrow (x, y) = (\pm a, \pm a)$

當 $\mu = -\frac{2}{3}\lambda$ 時, $x = y \Rightarrow x^2 = y^2 = \frac{a^2}{3} \Rightarrow (x, y) = (\pm \frac{a}{\sqrt{3}}, \pm \frac{a}{\sqrt{3}})$

$$\Rightarrow z = \frac{a}{3} \Rightarrow (\mu, \lambda) = (-\frac{2}{3}, \frac{1}{3})$$

同時, 考慮方程組 $\begin{cases} x^2 + y^2 = 2az \\ x^2 + y^2 + xy = a^2 \end{cases} \Leftrightarrow \begin{cases} x^2 + y^2 = 2az \\ xy = a^2 - 2az \end{cases} \Rightarrow \begin{cases} x^2 + y^2 = 2az \\ x^2 y^2 = (a^2 - 2az)^2 \end{cases}$

設 x^2, y^2 是一元二次方程 $t^2 - (2az)t + (a^2 - 2az)^2 = 0$ 的兩根

$$\Delta = (2az)^2 - 4(a^2 - 2az)^2$$

$$= 4a^2(z^2 - (a - 2z)^2)$$

$$= 4a(3z - a)(a - z) \geq 0 \Leftrightarrow \frac{a}{3} \leq z \leq a$$

因此, 所求最大距離是 a , 最短距離是 $\frac{a}{3}$

4. 設 $f(x) = x^p$

$$f''(x) = p(p-1)x^{p-2}$$

當 $p > 1$, $x \geq 0$ 時, $f''(x) \geq 0$

根據 Jensen 不等式知

$$\sum_{i=1}^n \frac{1}{n} f(a_i) \geq f\left(\frac{\sum_{i=1}^n a_i}{n}\right) \Leftrightarrow \sum_{i=1}^n \frac{a_i^p}{n} \geq \left(\frac{\sum_{i=1}^n a_i}{n}\right)^p$$

$$\Leftrightarrow \left(\frac{\sum_{i=1}^n a_i^p}{n}\right)^{\frac{1}{p}} \geq \frac{\sum_{i=1}^n a_i}{n} \quad (\text{當且僅當 } a_1 = a_2 = \dots = a_n \text{ 時取等號})$$